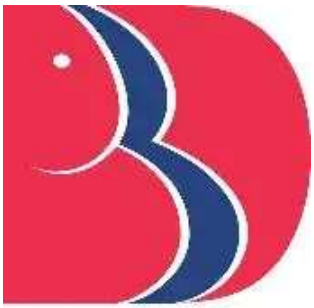
Babu Banarasi Das University



DESCRIPTIVE ANALYTICS (BCADSN23201)

TechNest Data Analytics Report

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TechNest Data Analytics

Dataset used: TechNest Sales Dataset, an electronics store with 60 orders in 2025. Columns: Order ID, Order Date, Region, Store Name, Category, Product, Quantity, Unit Price and Sales.

IBM COGNOS ANALYTICS

Practical 1: Data Source Connection & Import

Objective: Upload and import the TechNest sales dataset into IBM Cognos Analytics to prepare it for analysis.
Materials Required: IBM Cognos Analytics access, TechNest_Sales_Dataset.xlsx, a personal content folder to save into.

Step-by-Step Procedure:

1. Opened IBM Cognos Analytics and uploaded **TechNest_Sales_Dataset.xlsx** as a data module.
2. Selected the **Sales** sheet and checked that all 9 columns loaded with the correct data types.
3. Created a new **List** report on this data.
4. Dragged Order ID, Order Date, Region, Store Name, Category, Product, Quantity, Unit Price and Sales onto the canvas.
5. Clicked **Run** to preview the live data.
6. Saved the report as "TechNest - First Sales Report" in the personal content folder.

Expected Outcome: The report shows live TechNest rows (first 10 of 60 shown below), with correctly labelled columns.

Order ID	Order Date	Region	Store Name	Category	Product	Qty	Unit Price	Sales
ORD1001	01-01-2025	North	Lucknow Mall	Accessories	Wireless Earbuds	1	2,500	2,500
ORD1002	11-01-2025	East	Kolkata Square	Accessories	Smart Watch	2	4,200	8,400
ORD1003	15-01-2025	South	Chennai Hub	Accessories	Power Bank	3	1,400	4,200
ORD1004	02-02-2025	South	Bengaluru Tech Park	Mobiles	Feature Phone	3	2,200	6,600
ORD1005	06-02-2025	South	Chennai Hub	Laptops	Student Laptop	1	28,000	28,000
ORD1006	07-02-2025	East	Kolkata Square	Laptops	Student Laptop	1	28,000	28,000
ORD1007	09-02-2025	South	Bengaluru Tech Park	Laptops	Laptop Pro 15	3	55,000	1,65,000
ORD1008	11-02-2025	East	Patna Plaza	Accessories	Power Bank	3	1,400	4,200
ORD1009	13-02-2025	North	Lucknow Mall	Laptops	Laptop Air 13	3	42,000	1,26,000
ORD1010	22-02-2025	West	Pune Point	Laptops	Laptop Pro 15	1	55,000	55,000

Practical 2: Build a Grouped TechNest Sales List Report

Related Topic: Create List Reports
Objective: Build a list report, group it, format its columns and add a summarizing footer.
Materials Required: IBM Cognos Analytics access, TechNest dataset, report from Practical 1.

Step-by-Step Procedure:

1. Create a new List report on the TechNest data.
2. Add Region, Store Name, Quantity and Sales as columns.
3. Group the report by Region, then by Store Name.
4. Format Sales as currency and right-align the numeric columns.
5. Add a footer using the Total (sum) aggregate.
6. Insert a page header with the title and run date, then run and save.

Expected Outcome: Rows are grouped by Region and Store Name, numbers are right-aligned and the group totals add up to the overall total.

Region	Store Name	Quantity	Sales (Rs.)
North	Delhi Central	20	2,98,600
North	Lucknow Mall	37	5,14,700
North - Total		**57**	**8,13,300**
South	Bengaluru Tech Park	17	4,29,700
South	Chennai Hub	11	1,55,700
South - Total		**28**	**5,85,400**
East	Kolkata Square	20	2,44,000
East	Patna Plaza	16	1,35,400
East - Total		**36**	**3,79,400**
West	Mumbai Arcade	14	62,800
West	Pune Point	16	1,78,200
West - Total		**30**	**2,41,000**
Overall - Total		**151**	**20,19,100**

Practical 3: Filter TechNest Sales by Region and Date

Related Topic: Focus Reports Using Filters
Objective: Filter the report to one region and one period, then make it reusable with a prompt.
Materials Required: IBM Cognos Analytics access, grouped list report from Practical 2.

Step-by-Step Procedure:

1. Open the TechNest list report from Practical 2.
2. Add a simple filter: Region = "North".
3. Add a detail filter: Order Date between 01-Jan-2025 and 31-Mar-2025.
4. Combine both conditions with AND.
5. Run the report and confirm the row count drops.
6. Convert the Region filter into a prompted filter.

Expected Outcome: Only North-region rows of the first quarter remain (5 of 60 rows).

Order ID	Order Date	Region	Store Name	Category	Quantity	Sales (Rs.)
ORD1001	01-01-2025	North	Lucknow Mall	Accessories	1	2,500
ORD1009	13-02-2025	North	Lucknow Mall	Laptops	3	1,26,000
ORD1011	01-03-2025	North	Lucknow Mall	Laptops	3	1,65,000
ORD1015	18-03-2025	North	Lucknow Mall	Mobiles	1	2,200
ORD1017	25-03-2025	North	Delhi Central	Mobiles	2	36,000
Total					**10**	**3,31,700**

Practical 4: Build a TechNest Region x Category Crosstab

Related Topic: Cross-Tabular Reporting
Objective: Compare sales across regions and product categories with a crosstab.
Materials Required: IBM Cognos Analytics access, TechNest dataset.

Step-by-Step Procedure:

1. Create a new **Crosstab** report on the TechNest data.
2. Drag **Region** to the rows zone.
3. Drag **Category** to the columns zone.
4. Drag **Sales** into the cell zone.
5. Run the report to preview the matrix.

Expected Outcome: A grid with regions on rows, categories on columns and the sales value at each intersection.

Sales	Mobiles	Laptops	Accessories
North	1,12,400	5,99,000	1,01,900
South	76,500	4,98,000	10,900
East	42,600	2,77,000	59,800
West	85,500	1,10,000	45,500

Practical 5: Add Totals and Summarization to Crosstab

Related Topic: Advanced Crosstab Formatting and Summary Functions
Objective: Add row and column totals to the crosstab from Practical 4.
Materials Required: IBM Cognos Analytics access, crosstab report from Practical 4.

Step-by-Step Procedure:

1. Open the Practical 4 crosstab in authoring mode.
2. Select the row and column headers.
3. Click the **Summarize** icon in the toolbar.
4. Choose **Total** for rows and columns.
5. Run the report to see the totals.

Expected Outcome: Row totals, column totals and the grand total (Rs. 20,19,100) appear next to the cell values.

Sales	Mobiles	Laptops	Accessories	Total
North	1,12,400	5,99,000	1,01,900	**8,13,300**
South	76,500	4,98,000	10,900	**5,85,400**
East	42,600	2,77,000	59,800	**3,79,400**
West	85,500	1,10,000	45,500	**2,41,000**
Total	**3,17,000**	**14,84,000**	**2,18,100**	**20,19,100**

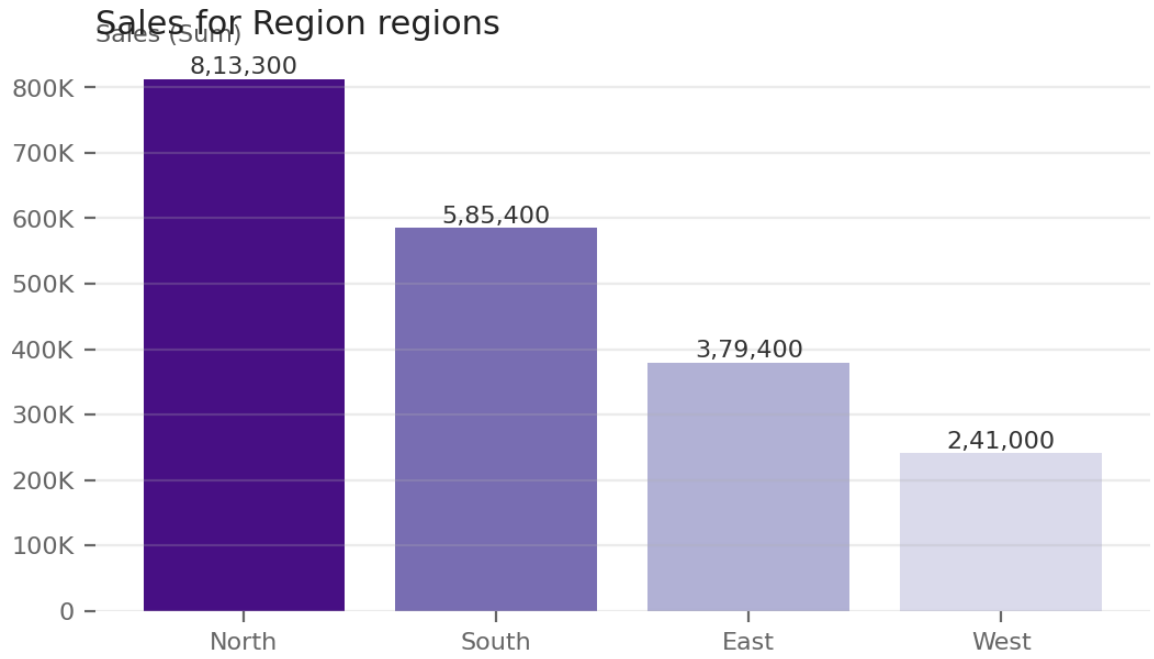
Practical 6: Create a Visual Chart in IBM Cognos Analytics

Related Topic: Data Visualization and Dashboarding
Objective: Turn the sales data into a chart that shows regional performance.
Materials Required: IBM Cognos Analytics access, TechNest dataset.

Step-by-Step Procedure:

1. Open Cognos Analytics and click **New (+) > Dashboard**.
2. Select the TechNest data module as the source.
3. Choose a Column chart.
4. Drag **Region** to the category axis and **Sales** to the value axis.
5. Preview the dashboard.

Expected Outcome: A column chart showing total sales for North, South, East and West.



Conclusion & Summary (Cognos)

Using IBM Cognos Analytics and the TechNest dataset, I built grouped list reports, applied filters and prompts, created crosstabs with totals and visualised sales by region.

POWER BI DESKTOP LAB PRACTICALS: TECHNEST ANALYTICS

Overview: Microsoft Power BI Desktop is used to build dashboards, model data, write DAX measures and create interactive reports from the TechNest dataset.
Setup: Open Power BI Desktop, import TechNest_Sales_Dataset.xlsx through Get Data > Excel, check data types in Power Query Editor and click Load.

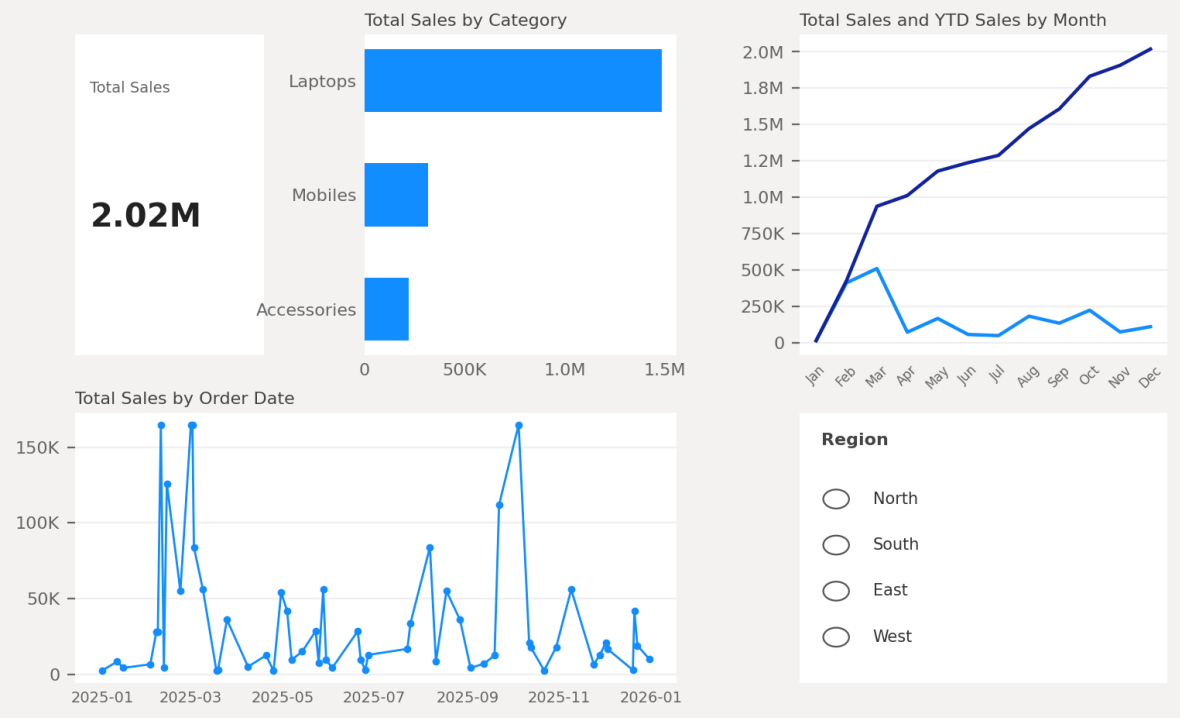
Power BI Practical 1: Data Import and Basic Sales Dashboard

Related Topic: Data Connectivity & Visualization
Objective: Import the TechNest dataset and build a basic dashboard with a column chart and KPI card.
Materials Required: Power BI Desktop, TechNest_Sales_Dataset.xlsx.

Step-by-Step Procedure:

1. Open Power BI Desktop and click **Get Data > Excel**.
2. Select the **Sales** sheet and click **Load**.
3. Choose a **Clustered Column Chart**.
4. Drag Region to the X-axis and Sales to the Y-axis.
5. Add a **Card** visual for total Sales.

Expected Outcome: A report page with sales by region and a card showing total sales of 2.02M.



Power BI Practical 2: DAX Measures & Slicers

Related Topic: Data Analysis Expressions (DAX) and Interactive Filtering
Objective: Create DAX measures (Total Sales and YTD Sales) and add a slicer.
Materials Required: Power BI file from Practical 1.

Step-by-Step Procedure:

1. Right-click the Sales table and choose **New Measure**.
2. Write: Total Sales = SUM(Sales[Sales]).
3. Create: YTD Sales = TOTALYTD([Total Sales], Sales[Order Date]).
4. Add a **Slicer** with Region or Category.
5. Test the slicer on the visuals.

Expected Outcome: Working DAX measures; the table below shows Total Sales and YTD Sales for each month.

Month	Total Sales (Rs.)	YTD Sales (Rs.)
Jan	15,100	15,100
Feb	4,12,800	4,27,900
Mar	5,11,000	9,38,900
Apr	73,700	10,12,600
May	1,68,000	11,80,600
Jun	57,800	12,38,400
Jul	50,300	12,88,700
Aug	1,83,400	14,72,100
Sep	1,35,700	16,07,800
Oct	2,24,500	18,32,300
Nov	75,200	19,07,500
Dec	1,11,600	20,19,100

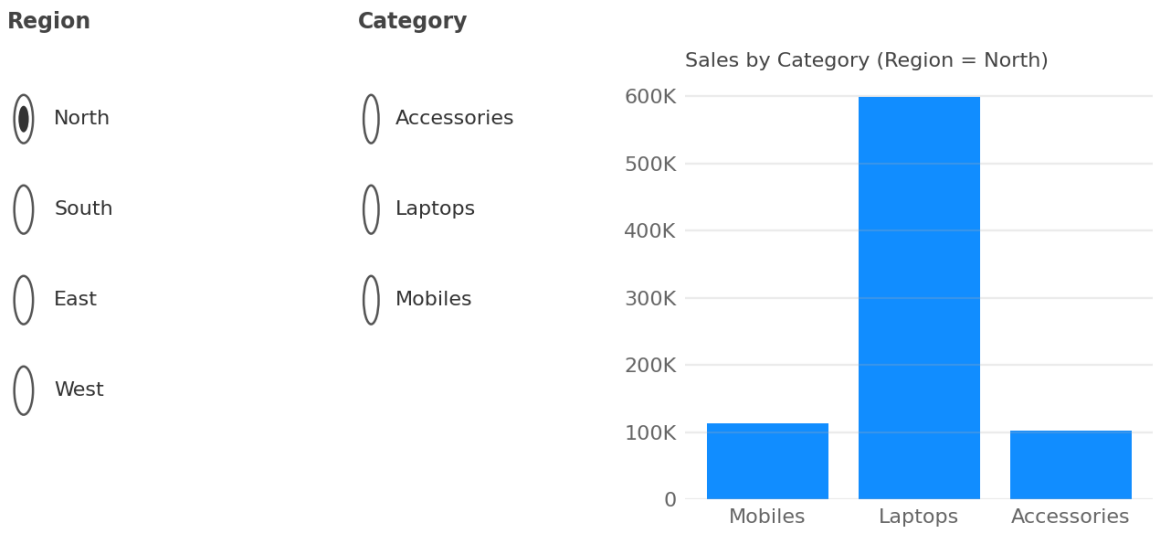
Power BI Practical 3: Interactive Slicers and Filters

Related Topic: Report Filtering and Dynamic Interactivity
Objective: Add Region and Category slicers to explore subsets of the data.
Materials Required: Power BI Desktop with TechNest dataset loaded.

Step-by-Step Procedure:

1. Open the report canvas.
2. Click the **Slicer** icon in the Visualizations pane.
3. Drag **Region** into the slicer and add another for **Category**.
4. Format the slicers as list or dropdown.
5. Select different values and watch the visuals change.

Expected Outcome: Slicers that filter the whole dashboard instantly (example: Region = North).



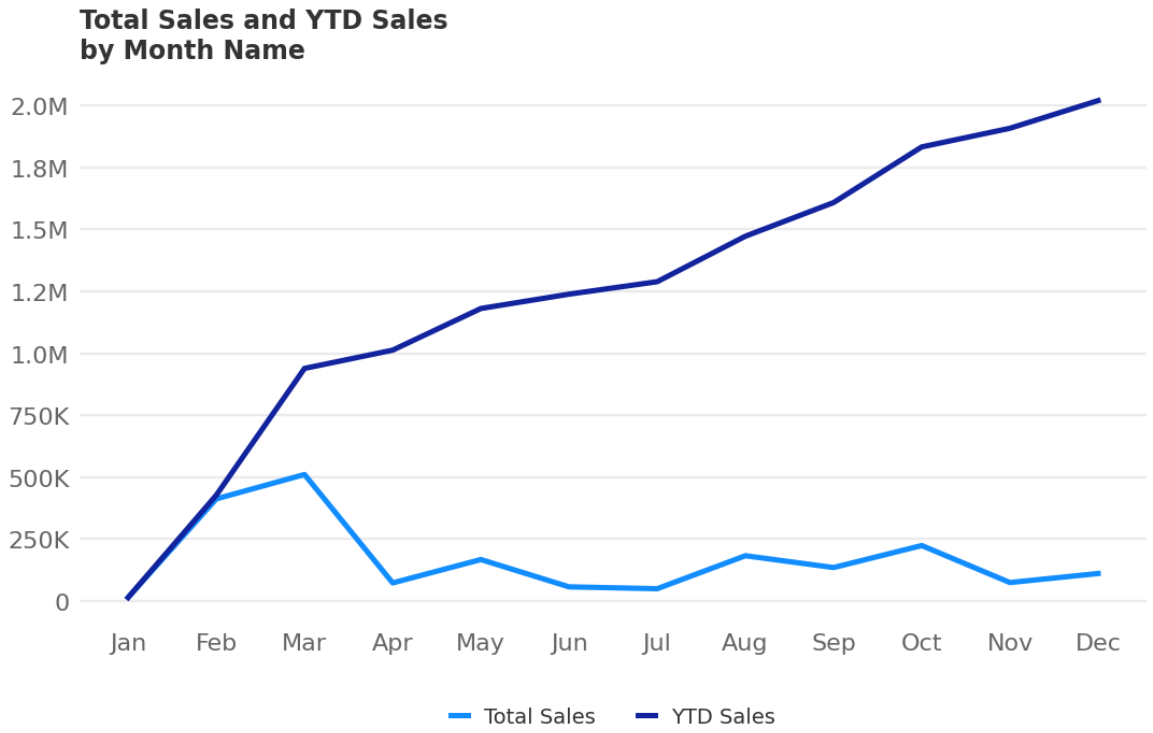
Power BI Practical 4: Time Intelligence & YTD Trends

Related Topic: Trend Lines
Objective: Compare monthly sales with the Year-to-Date total.
Materials Required: Power BI Desktop with TechNest data and the measures from Practical 2.

Step-by-Step Procedure:

1. Select a **Line Chart** visual.
2. Drag **Order Date** (Month) to the X-axis.
3. Drag **Total Sales** and **YTD Sales** to the Y-axis.
4. Use different colours and turn on data markers.
5. Save the report.

Expected Outcome: A line chart comparing monthly sales with cumulative YTD sales through 2025.



Final Conclusion & Summary

Through these 10 practicals on IBM Cognos Analytics (1 to 6) and Power BI Desktop (1 to 4), I used the TechNest dataset to practise data import, grouped reports, filters, crosstabs, totals, charts, slicers and DAX based time analysis for business decision-making.